

Today's agenda

1) UMAP

2) Housing Price Prediction Problem

$$- \sum_{i=1}^m \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]$$

↘ CE loss
log-loss
for binary classification

1 7



If we fix the random state for UMAP, can we reproduce the results?

1960	1941	1942
↓	↓	↓
1	2	3
1.5		2.5

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 7 \\ 2 & 4 & 6 \end{bmatrix}$$

rank = 2

$\hookrightarrow$ inv 3×3

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 7 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

$X \quad Y \quad Z$

$$\begin{cases} x + 2y + 3z = 3 \\ 2x + y + 7z = 4 \end{cases}$$

$$\rightarrow 2x + 4y + 6z = 6$$

$$\begin{matrix} & x & y & z \\ \begin{matrix} x \\ y \\ z \end{matrix} & \begin{bmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_y^2 & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_z^2 \end{bmatrix} \end{matrix} \sim \begin{matrix} x, y, z \\ (x - \mu_x)(y - \mu_y) \end{matrix}$$

$$\sigma_x = \sqrt{\frac{1}{n} \sum (x - \mu_x)^2}$$

$$\sigma_y = \sqrt{\frac{1}{n} \sum (y - \mu_y)^2}$$

✓